

JEE MAIN REVISION MASTER: MATRICES AND DETERMINANTS 2026 9 QUESTIONS 15% WEIGHTAGE $\uparrow$ 12.53%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 65 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Matrices And Determinants - Core Application	65 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Matrices And Determinants - Core Application (Very High Yield — 65 PYQs)

Core Principle: Key problem pattern in Kcet Matrices And Determinants.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area of the triangle with vertices (3,8),(-4,2) and (5,1) is $(P / 4)$, then the value of P is..." [Ans: C]
- [2026 Shift] Q#1 (MCQ): "Consider the following statements: Statement I: If A is a non-singular matrix, then A^{-1} exists. Statement II: If A and B are symmetric ..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Which of the following is correct?..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area of the triangle with vertices $(3,8)$, $(-4,2)$ and $(5,1)$ is $(P/4)$, then the value of P is

- (A) $(61/2)$
- (B) $(2/61)$
- (C) 122 **[Correct]**
- (D) $(1/122)$

Correct Answer: C

Solution Analysis: Using the area formula for a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) ,
$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$
 Here the vertices are $(3,8)$, $(-4,2)$ and $(5,1)$. So,
$$\text{Area} = \frac{1}{2} |3(2-1) + (-4)(1-8) + 5(8-2)| = \frac{1}{2} |3(1) + (-4)(-7) + 5(6)| = \frac{1}{2} |3 + 28 + 30| = \frac{1}{2} \times 61 = \frac{61}{2}$$
 Given that the area is $(P/4)$, $(P/4) = (61/2)$ Multiplying both sides by 4, $P = 122$ So, the correct option is 122 Option C.